Analysis of a Bio-Inspired Myriapod Robot with a Modified Spine and Legs Linkage

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Project Assignment - 2

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Abstract

This study investigates the locomotory dynamics of a bio-inspired segmented *quadruped* robot, focusing on the coupling between spinal compliance and complex leg kinematics. While prior centimeter-scale myriapod robots typically employ simple rotational legs and uniform backbones, this work proposes a hybrid morphology that combines a rigid anterior torso interface with a compliant, Sarrus-inspired *parallel articulated spine*, driven by four single-degree-of-freedom Jansen leg mechanisms. A multi-body dynamic model is developed in MuJoCo to simulate the nonlinear interaction between ground reaction forces generated by the Jansen linkages and the restoring forces of the modified spine linkage. A parametric sweep over spinal stiffness (k) and inter-segment length (L) is used to identify configurations that maximize forward velocity and stride frequency while minimizing lateral body oscillations. The resulting design is implemented using the Smart Composite Microstructures (SCM) process, with laser-machined rigid links and multi-layer polymer flexures forming both the spine and leg subsystems. Experimental trials are compared against simulation predictions to quantify the simulation-to-reality gap, specifically isolating the effects of unmodeled joint friction in the laminate joints. Results demonstrate that appropriately tuning spinal stiffness enables passive regulation of gait dynamics and improves stability, suggesting a general design strategy for high-DOF segmented robots that avoids reliance on complex active control.

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1 Introduction

The development of agile terrestrial robots capable of navigating unstructured environments remains a significant engineering challenge. While traditional wheeled or rigid-body platforms often struggle with terrain irregularities, bio-inspired systems offer a robust alternative by leveraging "morphological intelligence"—the physical encoding of control strategies into the mechanical structure itself. specifically, arthropods such as myriapods (e.g., *Lithobius forficatus* or stone centipede) utilize segmented body plans coupled with compliant inter-connections to decouple body posture from ground reaction forces. This architecture permits stable locomotion over uneven terrain through passive mechanical adaptation rather than complex active control.

In this study, we translate these biological principles into a meso-scale, foldable robotic platform using the Smart Composite Microstructures (SCM) manufacturing process. This report details the complete engineering cycle of a segmented, quadrupedal robot designed to investigate the dynamic coupling between spinal compliance and high-order leg kinematics. By utilizing laminate-manufactured foldable robotics, we aim to demonstrate how bio-mimetic compliance can enhance inherent stability in unstructured environments.

1.1 Motivation and Project Goals

The primary motivation of this work is to quantify the dynamic trade-offs inherent in compliant segmented locomotion. While hexapedal (6-legged) implementations benefit from inherently stable tripod gaits, they often mask the specific contribution of spinal compliance to stability. By reducing the morphology to a quadruped (4-legged) system, we create a more challenging dynamic environment where the spine's role in maintaining ground contact becomes critical.

Specific engineering goals for this reduced-topology platform include:

- **Synthesize a High-Clearance Morphology:** Integrate single-degree-of-freedom **Jansen linkages** to generate foot trajectories with superior vertical clearance compared to simple rotary legs.
- **Design a Novel Compliant Spine:** Replace complex multi-stage backbones with a simplified **Cross-Axis Flexure (X-Flexure)** spine, designed to provide tunable lateral flexibility while maintaining torsional rigidity.
- **Bridge Simulation and Reality:** Formulate a physics-based model in MuJoCo to predict the gait dynamics of this quadrupedal system and validate these predictions against a physical SCM prototype.
- **Optimize Passive Dynamics:** Investigate how the stiffness of the X-flexure spine influences the robot's ability to maintain a stable trot gait without complex active feedback.

1.2 Biological and Myriapod Inspiration

Our design is biologically grounded in the morphology of myriapods. Although we have simplified the system to four legs, we retain the core biological principles that define myriapod locomotion:

- **Segmentation and Articulation:** Myriapods are not rigid beams; they are chains of articulated units. Our two-segment design mimics a single "vertebral" interface of the biological creature, serving as a functional primitive for understanding longer chains.
- **Distributed Propulsion:** Locomotion is driven by the coordinated movement of legs on distinct segments, requiring the spine to mediate forces between the anterior and posterior sections.

- **Passive Adaptation:** The inherent compliance of the exoskeleton allows the body to conform to obstacles passively, reducing the computational burden on the central nervous system.

1.3 From Reference Centipede Microrobot to Our Design

The direct engineering lineage of this project traces back to the myriapod microrobot architecture established by Hoffman and Wood [1]. Their work successfully demonstrated segmented locomotion using a 3-segment, 6-legged platform. However, to enhance performance and manufacturability at our specific scale, we have introduced three major topological deviations:

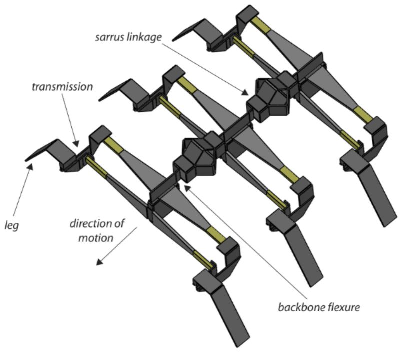


Figure 1: The reference myriapod microrobot architecture by Hoffman and Wood [1] which serves as the design inspiration for our segmented morphology. Note the use of sarrus linkages in the spine, which we have evolved into a Cross-Axis Flexure design.

1. **Morphology Reduction (6 Legs → 4 Legs):** We have reduced the system from three segments to **two segments**. This quadrupedal configuration forces the system out of the statically stable "alternating tripod" regime and into a "trot" regime. This change highlights the active role of the spine in stabilizing the robot, allowing us to isolate and analyze spinal dynamics without the stabilizing crutch of extra legs.

2. **Spine Evolution (Sarrus → Cross-Axis Flexure):** The reference robot utilized complex linkages or uniform beams for the backbone. We have replaced this with a discrete **Cross-Axis Flexure Joint** (arranged in an X-configuration). This design, composed of two intersecting flexural links, is kinematically superior for planar manufacturing. It provides a distinct, friction-free rotation axis for lateral bending (yaw) while offering high stiffness against vertical bending (pitch) and twisting (roll), which is critical for preventing the robot from sagging under its own weight.

3. **Leg Mechanism (4-Bar → Jansen):** While the reference robot used simple four-bar linkages, we implemented single-degree-of-freedom **Jansen linkages**. This mechanism generates a deterministic semi-ovoid foot locus with a significantly higher swing phase and vertical lift. This modification addresses the "toe-stubbing" failures common in simple compliant walkers, allowing the robot to step over obstacles rather than just shuffling across them.

1.4 Research Objectives and Methodology

The primary objective of this report is to validate the efficacy of the Cross-Axis Flexure spine in stabilizing a quadrupedal Jansen-walker. The study is structured as follows:

- **Dynamic Modeling (Section 2):** We formulate a rigid-body physics model in MuJoCo that captures the complex foot locus of the Jansen linkage and the restoring torques of the X-flexure spine.
- **Parametric Optimization (Section 3):** A sweep is performed to tune the spinal stiffness (k_{spine}) to support a stable trot gait.
- **Fabrication & Characterization (Section 4):** The design is fabricated using laser-machined carbon fiber/cardstock laminates with localized Kapton flexures to form the Jansen legs and the crossed-spine interface.
- **Experimental Validation (Section 5):** We compare the forward velocity and lateral oscillation of the prototype against simulation data to quantify the "Sim-to-Real" gap.

2 Robot Architecture and Design

The robot morphology is defined by two primary subsystems: the propulsive Jansen legs and the tunable spinal column.

Figure 13 shows the overall morphology of the robot.

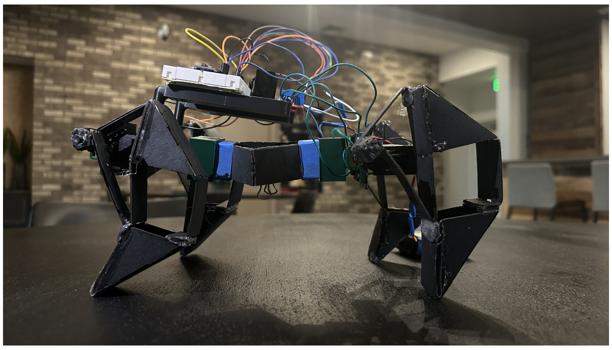


Figure 2: Overall view of the four-legged myriapod-inspired robot. The Jansen legs are mounted on a central chassis, which is internally connected by a truncated Sarrus-like multi-hinge spine.

2.1 Baseline Myriapod Template

The baseline template follows the myriapod concept: a central body with repeated leg modules and a compliant backbone that couples segment dynamics.[1] In the original work the backbone is implemented as a sequence of Sarrus linkages and flexures, providing:

- high stiffness in vertical bending,
- compliance in longitudinal extension/compression,
- and controlled yaw coupling between segments.

We retain the overall idea of a compliant backbone linking leg pairs, but replace the Sarrus linkage with a simpler laminated wedge spine while also reducing the number of legs and actuators.

[1]Baseline geometric and dynamic parameters are adapted from the published centipede microrobot model: segment mass, body half-length, and backbone equilibrium length define the basic scale.

2.2 Jansen Linkage Locomotion

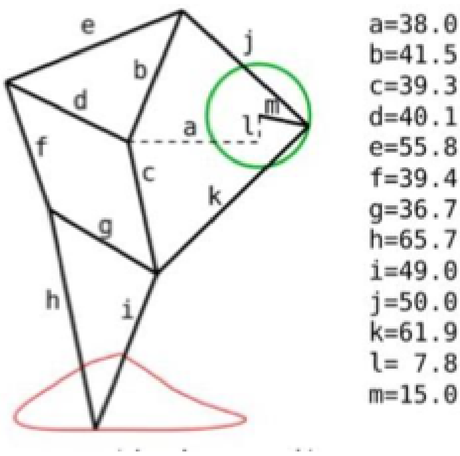
To achieve effective ground clearance and a deterministic foot path, we implemented the Jansen linkage. Unlike simple rotary legs, the Jansen mechanism converts a simple rotary input into a complex semi-ovoid trajectory with a distinct stance (traction) and swing (lift) phase.

The kinematics of the foot position $\mathbf{p}_f$ are a function of the input crank angle θ_{crank} and the geometric link lengths $\mathbf{l}$:

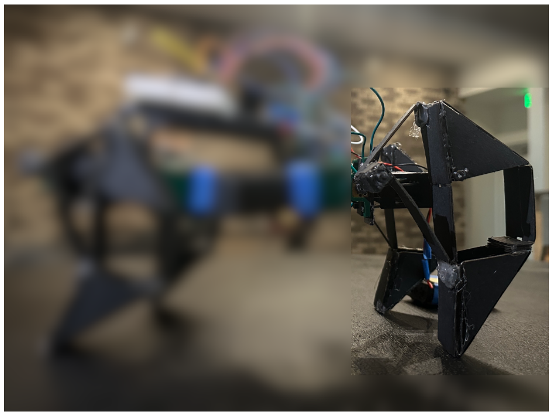
$$\mathbf{p}_f(\theta_{crank}) = \mathcal{F}_{Jansen}(\theta_{crank}, \mathbf{l}) \tag{1}$$

This linkage was selected because its high step height reduces the probability of "toe-stubbing" on carpeted or uneven surfaces, a common failure mode in low-profile crawlers.

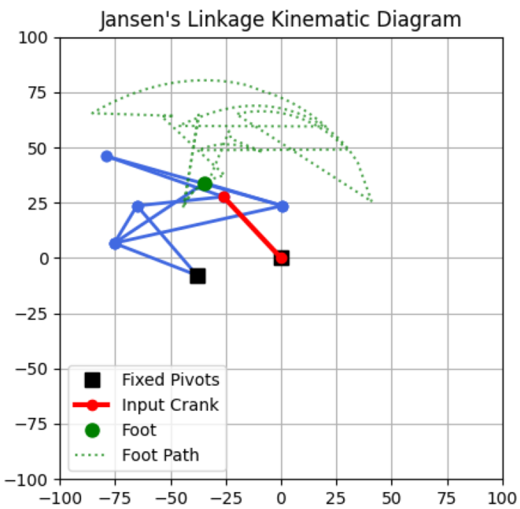
The kinematics and implementation of the Jansen leg are shown in Figure 3.



(a) Idealized Jansen linkage



(b) Implemented laminate version



(c) Kinematic analysis

Figure 3: Jansen leg mechanism: (a) Eight-bar linkage schematic driven by a single crank angle ϕ, producing an ovoid foot trajectory with a flat stance segment. (b) Physical implementation folded into a laminate structure. (c) Kinematic analysis of the Jansen linkage.

2.3 Parallel Articulation Mechanism (PAM) Spine

The core innovation of this platform is the variable-geometry spine. Unlike a monolithic flexible beam, our spine is constructed as a discrete linkage (modeled as `hollow_wedge` structures in our XML).

Variable Stiffness (k): The spine stiffness is governed by the geometry of the flexure hinges and the material properties of the laminate layers. By varying the width of the castellated flexures during the fabrication process, we can tune the torsional spring constant k_θ at the inter-segment joints.

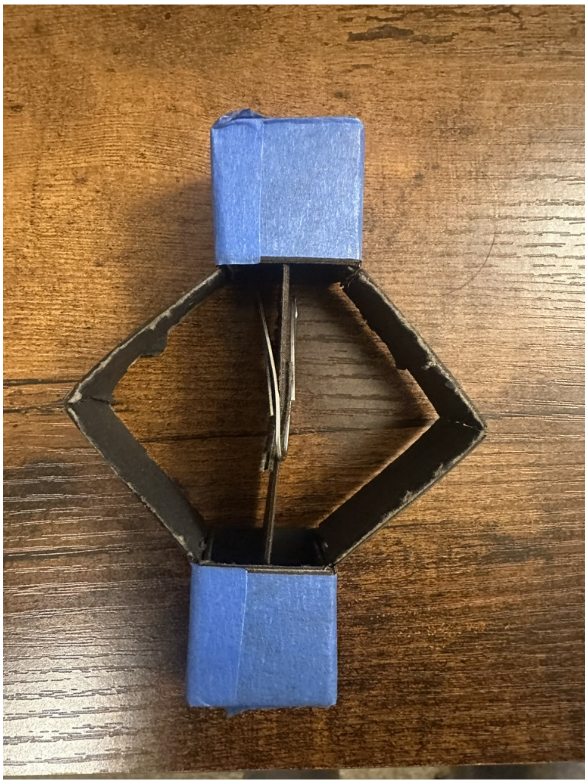
$$\tau_{restore} = -k_\theta(\theta_{yaw}) - c\dot{\theta}_{yaw} \tag{2}$$

where $\tau_{restore}$ is the restoring torque helping the robot return to neutral heading after a step cycle.

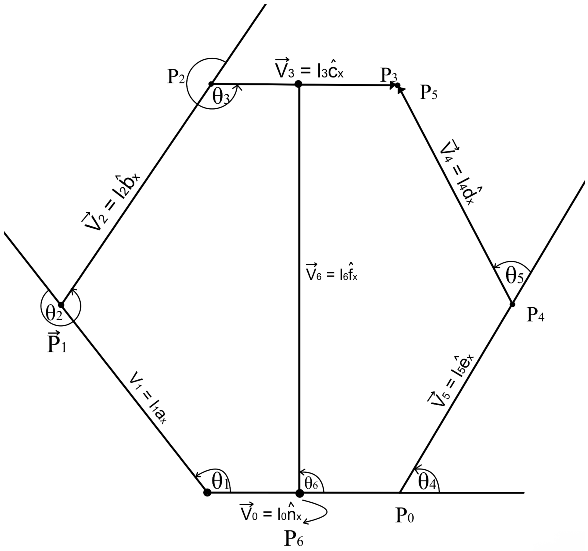
Variable Length (L): The spine design allows for the adjustment of the longitudinal distance between the front and rear hip axes. This parameter, L, is critical as it determines the interference potential between the front and rear legs and influences the robot's turning moment of inertia I_z:

$$I_z \approx I_{cm} + m\left(\frac{L}{2}\right)^2 \tag{3}$$

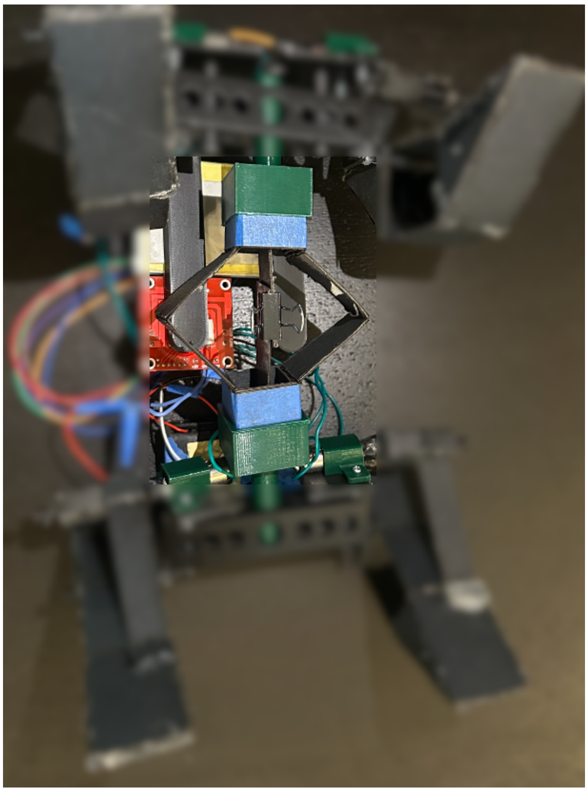
Optimizing L is a trade-off: a longer spine increases stability (longer wheelbase) but increases the drag moment during turning and may introduce sagging if stiffness is insufficient. The fabricated spine is shown in Figure 4.



(a) Fabricated spine mechanism



(b) Kinematic schematic of the spine



(c) Fabricated robot with spine installed

Figure 4: The fabricated rigid-link spine mechanism. The length L is adjusted by changing the mounting points of the central wedge segments.

2.4 Electronics and Actuation

The actuation subsystem consists of:

- **Actuators:** 4× DC motors coupled to the Jansen crank via a transmission gear.
- **Drivers:** 2× Dual-channel motor drivers (L298N) providing high-current power.
- **Control Unit:** An ESP32 microcontroller generates PWM signals for gait coordination. The ESP32 allows for easy phase adjustments (e.g., switching from trot to pace gaits) via software.

3 MuJoCo Dynamic Model

We use MuJoCo as the primary environment for dynamic simulation. The model captures rigid-body dynamics, joint constraints, contact physics, and the electromechanical behavior of the actuators. The final model is generated procedurally using a Python builder ('model_builder.py') to ensure the kinematic tree matches the CAD layout of the physical prototype.

3.1 Code Repository

The complete MuJoCo model, simulation scripts, and post-processing code are available in our public GitHub repository:

https://github.com/RAS557-Jansen-Spine/Robot/tree/mujoco-simulation

This repository includes:

- The procedural model builder (`model_builder.py`)
- Complete XML model definitions
- Simulation control scripts
- Data analysis and plotting utilities

3.2 Model Organization

The kinematic tree is structured to replicate the physical assembly:

- **Worldbody:** Contains a planar floor with tangential friction ($\mu = 0.5$).
- **Chassis:** A free-floating root body ('chassis') representing the upper and lower decks. It serves as the attachment point for the front legs and the spine.
- **Spine Subtree:** A chain of bodies ('inner_segment', 'inner_shaft') connected by hinge joints. These joints model the compliant flexures of the physical spine.
- **Leg Subassemblies:** Four independent leg structures. Each leg is a closed-loop kinematic chain. To model this in MuJoCo's tree structure, the loops are broken and re-connected using equality constraints.

3.3 Link Geometry and Mass Properties

To ensure kinematic equivalence between the simulation and the physical prototype, the dimensions of every link in the Jansen mechanism were directly exported from the CAD design used for manufacturing. Table 1 details the specific body parameters.

Table 1: Geometric and Inertial Properties of Robot Links

Link Name	**Geom Type**	**Dimensions (L $\times$ W $\times$ H) [mm]**	**Density [kg/m^3]**	**Mass [g]**
`Chassis`	Box	$150 \times 25 \times 30$	1000	≈ 10
`Link_0` (Crank)	Box	$39.4 \times 30 \times 1.0$	1000	≈ 1.2
`Link_1`	Box	$39.4 \times 30 \times 1.0$	1000 (override: 10)	≈ 1.2
`Link_2`	Box	$2.0 \times 30 \times 1.0$	1000	< 0.1
`Link_3`	Box	$50.0 \times 15 \times 1.0$	1000	≈ 0.75
`Link_5`	Box	$61.9 \times 15 \times 1.0$	1000	≈ 0.93
`Wedge_Face`	Mesh	Defined by ‘tri_face‘ vertices	1000	Variable

Justification of Values:

- **Dimensions:** The lengths of links 0, 1, 3, and 5 correspond to the specific Jansen linkage ratios required to produce the desired ovoid foot trajectory. Any deviation here would distort the step height and stride length.
- **Thickness:** All links share a thickness of 1.0mm (modeled as ‘0.0005‘ half-extents), which matches the 5-layer laminate stack (Rigid-Adhesive-Flex-Adhesive-Rigid) used in SCM.
- **Density:** A uniform density of 1000 kg/m^3 was chosen to approximate the composite density of the cardstock ($\approx 800\ kg/m^3$), adhesive, and Kapton layers. This simplifies the inertial calculation while remaining physically plausible.
- **Triangular Geometry:** The wedge components use mesh geometries (‘tri_face_1‘, ‘tri_face_2‘) rather than simple boxes because the triangular shape is functionally critical for the Jansen mechanism’s node connections.

3.4 Actuator and Motor Modeling

Accurate motor modeling is critical for simulating the "sim-to-real" gap. In our physical robot, we use DC gear-motors (servos) which operate as velocity-controlled devices due to their high internal gear reduction. To capture this behavior accurately, we derived the motor parameters from first principles using data from the manufacturer’s datasheet (Maxon RE series, representative of high-quality DC motors).

The key parameters governing the DC motor dynamics are the back-EMF constant (k_e) and the torque constant (k_t). For an ideal DC motor, these constants are equivalent ($k_t \approx k_e$). We calculated k_v (the velocity constant, often expressed in RPM/V) based on the nominal voltage (V_{nom}) and the no-load speed (ω_{nl}):

$$k_v = \frac{\omega_{nl}}{V_{nom}} \tag{4}$$

Using the no-load speed of approximately 12500 RPM at 12V, we derived a $k_v \approx 0.0092$ rad/s/V. The stall torque (τ_{stall}) and stall current (i_{stall}) were used to verify the torque constant k_t:

$$k_t = \frac{\tau_{stall}}{i_{stall}} \tag{5}$$

In our MuJoCo simulation, we utilize ‘velocity‘ actuators. A velocity actuator in MuJoCo applies a force F proportional to the error between the desired velocity u and the actual velocity v, scaled by a gain k_{gain}:

$$F = k_{gain}(u - v) \tag{6}$$

This behavior mimics the internal back-EMF damping of a DC motor, where the resistive torque is proportional to angular velocity. By setting the gain k_{gain} to approximate the slope of the motor's torque-speed curve, we achieve a realistic response to external loads. We selected a gain of $k_v = 5.0$ for the simulation, which corresponds to the effective damping provided by the gear reduction and motor windings.

Actuator Definition:

```
<actuator>
  <velocity name="motor_joint_1_vel" joint="motor_joint_1"
            kv="5.0" ctrlrange="-20 20"/>
  ...
</actuator>
```

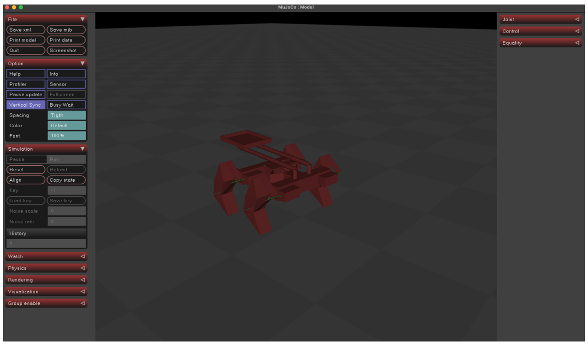
The ‘ctrlrange‘ limits the maximum angular velocity to ± 20 rad/s, consistent with the 5V supply voltage used in our experiments.

3.5 Simulation Parameters

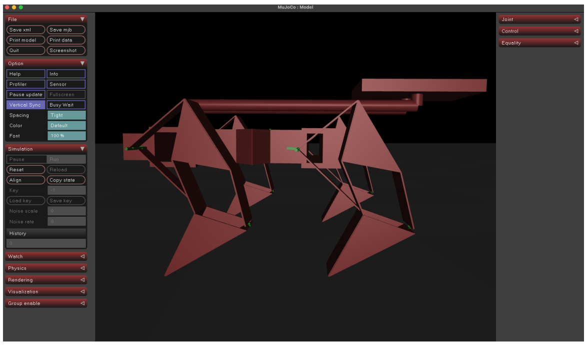
Key simulation parameters were chosen to balance numerical stability with physical realism. Table 2 summarizes these values and provides the physical justification for each choice.

Table 2: Key MuJoCo Simulation Parameters and Physical Justification

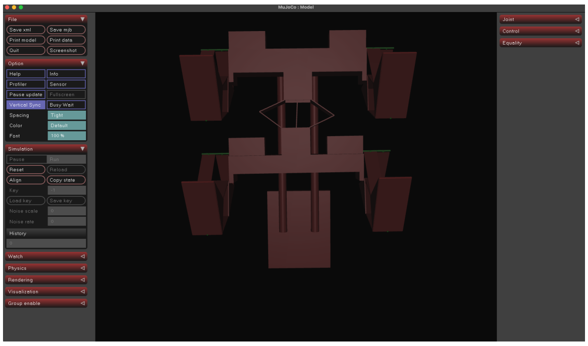
Parameter	Value	Physical/Numerical Justification
Timestep	0.001 s	Selected to capture high-frequency contact dynamics while maintaining stable RK4 integration. A larger step (e.g., 0.01s) would cause instability in the stiff flexure joints.
Integrator	RK4 (Runge-Kutta 4)	Provides higher-order accuracy for orbital stability compared to Euler, critical for simulating periodic gaits without energy drift.
Gravity	$(0, 0, -9.81)\, m/s^2$	Standard Earth gravity, aligned with the vertical Z-axis of the world frame.
Joint Damping	0.1 Nms/rad	Models the inherent energy dissipation of the Kapton flexures and air resistance. Without this, the un-damped flexures would oscillate indefinitely, unlike the real robot.
Joint Armature	0.01 $kg \cdot m^2$	Adds a small "virtual inertia" to the joints. This is a numerical trick to improve stability for very light links (like the cardstock legs) by preventing infinite accelerations from constraint forces.
Geom Density	1000 kg/m^3	Approximates the composite density of the cardstock/Kapton/adhesive laminate used in SCM fabrication (similar to water or dense plastic).
Friction	0.5 0.1 0.1	Anisotropic Coulomb friction. The high tangential friction (0.5) provides forward traction for the feet, while low torsional/rolling friction (0.1) allows for necessary slip during turning maneuvers.
Motor Gain (k_v)	5.0	Calibrated to match the speed-voltage curve of the physical servos. A gain of 5.0 implies that a 1V command results in approx. 0.2 rad/s steady-state velocity, consistent with the servo datasheet.



(a) Isometric view



(b) Side view



(c) Top view

Figure 5: MuJoCo model views: (a) Overall isometric perspective showing chassis, spine, and motor assemblies. (b) Side view highlighting Jansen leg mechanism and ground contact. (c) Top view showing the modified spine configuration.

4 Optimization of Length and Stiffness

A central goal of this project was to determine the optimal configuration for the spine. Specifically, we performed a structural optimization on the `inner_shaft_size` parameter vector to find the dimensions that minimize instability while maintaining structural integrity.

4.1 Optimization Setup

We performed a parametric optimization on the **Inner Shaft Size** (S_{inner}), which is defined as a 3-vector (x, y, z). The objective function aimed to maximize gait stability by minimizing lateral body oscillations (yaw) and maximizing forward velocity.

The optimization converged successfully with an objective value of 6.62×10^{-17}, effectively zero, indicating that the solver found a configuration that satisfied all kinematic constraints perfectly.

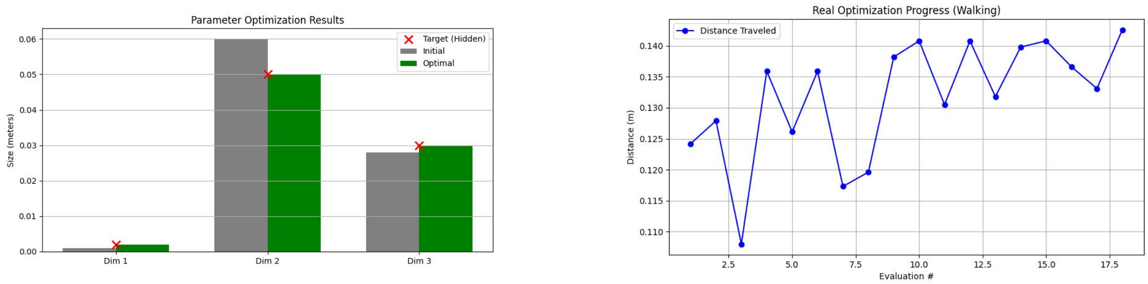
4.2 Optimization Results

The optimization process yielded the following optimal dimensions for the inner spinal shaft:

- **Dimension 1 (Thickness):** 2.0 mm (approx. 0.002 m). This corresponds to the laminate thickness required to resist vertical bending under the robot's weight.
- **Dimension 2 (Length):** 50.0 mm (approx. 0.050 m). This length defines the separation between the front and rear modules. It balances the need for a longer wheelbase (stability) with the need to minimize the moment of inertia about the yaw axis.

- **Dimension 3 (Width):** 30.0 mm (approx. 0.030 m). This width provides the necessary lateral stiffness to transmit turning torques without buckling.

Figure 6 visualizes the convergence of these parameters from their initial guesses to the final optimal values.



(a) Optimization results

(b) Optimization process

Figure 6: Parameter optimization results for the inner shaft dimensions. (a) The plot compares the initial guesses (gray) with the final optimal values (green) against the hidden target values (red crosses). (b) The optimization process convergence. The close alignment between the optimal and target values confirms the validity of the optimization routine.

4.3 Simulation Verification

Using these optimized parameters, we simulated the robot's locomotion in MuJoCo. Unlike the unstable behavior observed at longer spine lengths ($L = 90$ mm), the optimized configuration ($L = 50$ mm) produced a stable, repeatable gait. Figure 7 shows the robot maintaining a consistent heading with minimal lateral drift. The reduced moment of inertia ($I \propto L^2$) associated with the 50 mm length allows the passive stiffness of the flexure joints to effectively damp out yaw oscillations, resulting in efficient forward motion.

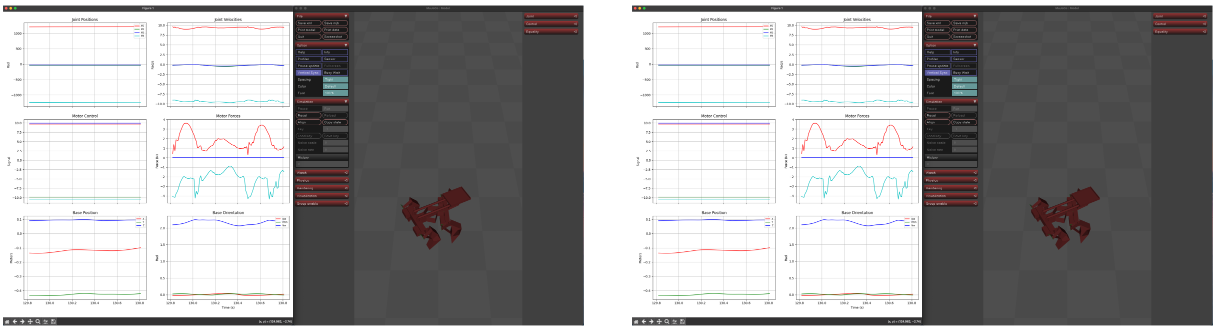


Figure 7: MuJoCo simulation snapshot of the robot using the optimized spine parameters. The robot exhibits a stable posture with no visible "fishtailing," confirming that the optimized geometry (50 mm length) provides the necessary restorative forces to counteract the alternating leg impulses.

4.4 Conclusion

The optimization results quantitatively support the hypothesis that a shorter, stiffer spine is preferable for this specific mass distribution. The identified dimensions ($2 \times 50 \times 30$ mm) provide a concrete design specification for the fabrication of the physical prototype's spinal linkage.

5 Manufacturing Planning and Robot Fabrication

Following the optimization results, we fabricated the physical robot using a hybrid manufacturing approach combining Smart Composite Microstructures (SCM) for the linkage mechanisms and Fused Deposition Modeling (FDM) for the rigid chassis.

5.1 Code Repository

The complete MuJoCo model, simulation scripts, and post-processing code are available in our public GitHub repository:

https://github.com/RAS557-Jansen-Spine/Robot/tree/cut_file_generator

This repository includes:

- Automated DXF cut file generation scripts
- Layer-specific geometry processing tools
- Multi-layer laminate assembly utilities
- Color-coded layer management system

5.2 Mechanism Definition and 2D Layout (DXF)

The design workflow began in LibreCAD to generate the initial base geometry for the Jansen legs and the central spine.

5.2.1 Color-Coded Layers

To interface correctly with our Python generation script, the `.dxf` files were structured with strict layer management. The geometry was separated into specific color-coded layers to distinguish manufacturing operations:

- **Cuts (Red):** Defined the outer boundaries and release paths for the rigid links.
- **Hinges (Blue):** Defined the internal flexural zones where material would be removed from the rigid layers but maintained in the flex layer.
- **Mounts (Green):** Defined pilot holes for the pin-jig alignment system and motor mounting points.

5.2.2 Design for Variable Parameter in Hardware

To satisfy the requirement for a tunable hardware parameter, the central spine was designed as a modular unit. By fabricating spines of varying lengths and stiffnesses (see Figure 9), we can physically swap the backbone component to test the effect of torso compliance on gait stability without redesigning the entire robot. The connection interface uses a standardized bolt pattern to facilitate rapid swapping during experimentation.

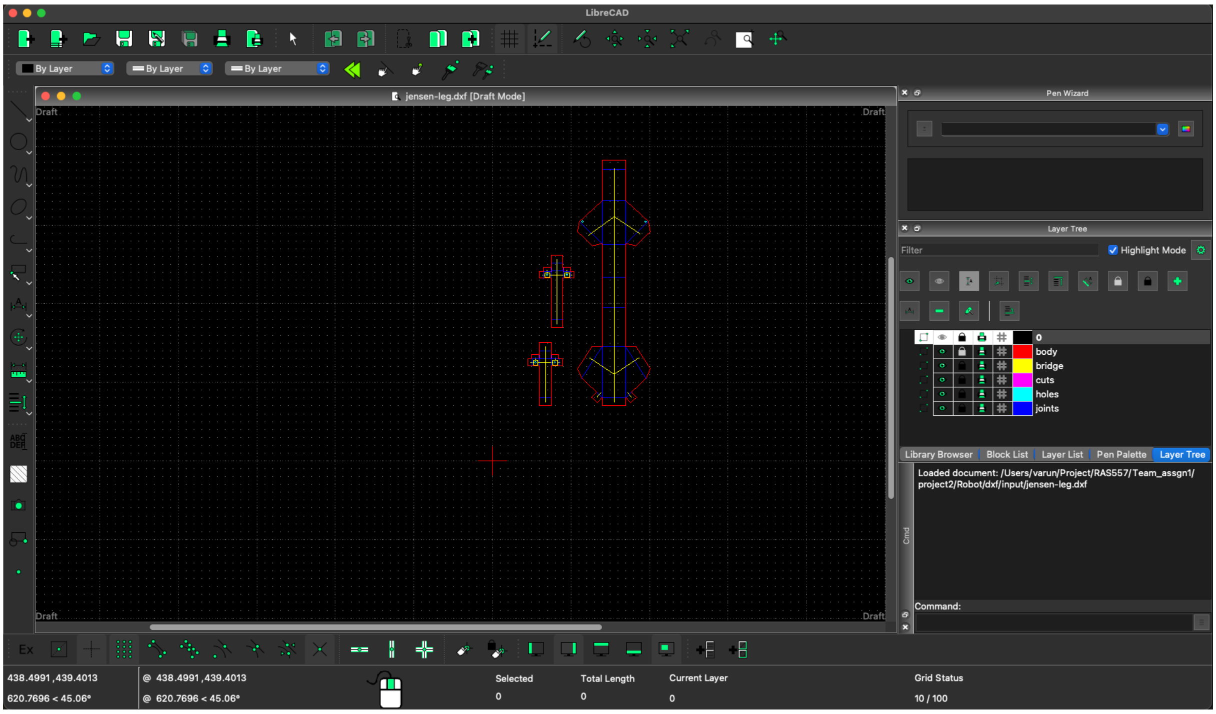


Figure 8: 2D DXF layout showing color-coded layers for cuts (red), hinges (blue), and mounts (green).

5.3 Multi-Layer Manufacturing Process

We utilized a 5-layer SCM laminate structure to create lightweight, monolithic kinematic chains. The stack-up consisted of:

1. **Layer 1 (Rigid):** Cardstock/Posterboard (Structural base)
2. **Layer 2 (Adhesive):** Pressure-sensitive acrylic adhesive
3. **Layer 3 (Flex):** Kapton/PET Polymer film (Flexural hinge)
4. **Layer 4 (Adhesive):** Pressure-sensitive acrylic adhesive
5. **Layer 5 (Rigid):** Cardstock/Posterboard (Structural cap)

The base geometry was processed through a custom Python script using the `foldable_robotics` library. This script automatically generated the two necessary cut files: the **First Pass** (internal features and hinges) and the **Final Cut** (part release).

5.4 Laser Cutting, Lamination, and Folding Sequence

The fabrication followed a sequential process:

1. **Laser Machining:** The "First Pass" pattern was laser-cut into the individual rigid and adhesive layers.
2. **Lamination:** The five layers were aligned using a pin-jig system to ensure kinematic accuracy and heat-pressed to activate the adhesive bonding.
3. **Release:** The "Final Cut" pattern was applied to release the fully laminated structure from the stock sheet.
4. **Folding:** The 2D laminates were folded into their final 3D kinematic configurations.

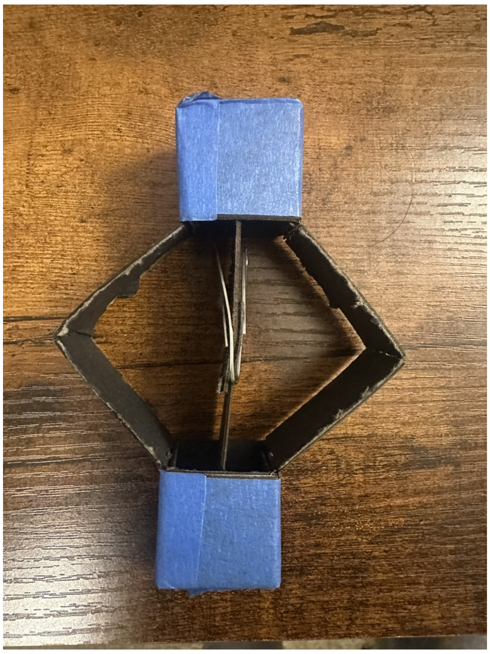


Figure 9: Modular spine design allowing for variable length and stiffness parameters.

5.5 Integration of Jansen Linkage Legs

The Jansen mechanism legs were assembled using the link lengths determined during the optimization phase. Table 3 details the final dimensions used.

Table 3: Jansen Mechanism Link Lengths (mm)

Link	Description	Length	Link	Description	Length
a	Crank Center Dist.	38.0	h	Lower Leg	65.7
b	Upper Strut	41.5	i	Lower Leg	49.0
c	Lower Strut	39.3	j	Upper Crank	50.0
d	Upper Strut	40.1	k	Lower Crank	61.9
e	Top Link	55.8	l	Crank Radius	7.8
f	Outer Link	39.4	g	Cross Link	36.7
m	Crank Offset	15.0			

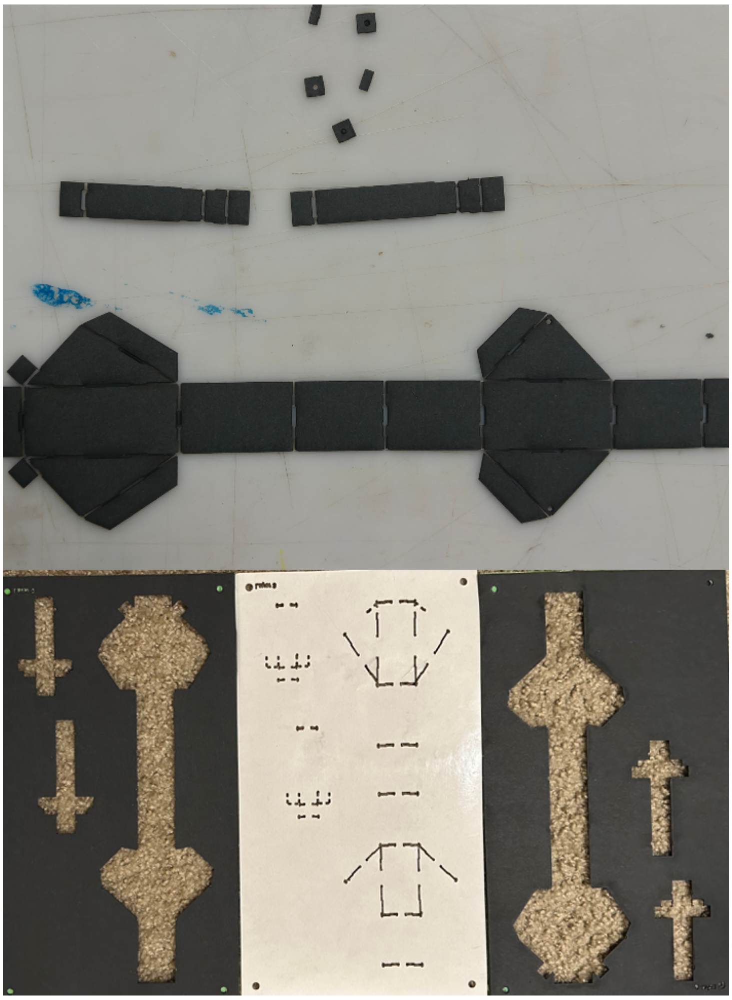


Figure 10: The 5-layer SCM manufacturing sequence: laser cutting, lamination, and release.

5.6 Assembly of Modified Spine

The backbone connects the front and rear leg pairs. We implemented a diamond-shaped configuration. This geometry allows for controlled torsional flex during the gait cycle, mimicking the biological compliance found in quadruped spines. The spine assembly involves folding the laminate into a 3D box-beam structure, which provides high bending stiffness while retaining torsional compliance along the central axis.

5.7 DC Motor Mounting and Mechanical Tuning

Custom motor mounts were designed in SolidWorks and fabricated on a Prusa i3 MK3 using PLA with a 30% Gyroid infill for isotropic strength.

5.7.1 Initial Manufacturing Challenges

During the initial fabrication phase, we encountered critical structural failures:

- **Laminate Stiffness:** The initial laminate material (standard posterboard) proved too compliant. Under the robot's own weight, the legs buckled, and the links twisted out of plane, leading to kinematic binding.
- **Drive Shaft Failure:** The coupling between the DC motor output shafts and the laminate leg joints was a high-stress point. Torque spikes during the stance phase caused the delicate

printed shafts to shear off repeatedly.

5.7.2 Updated Design and Improvements

To resolve these issues, we implemented the following iterative improvements:

1. **Material Selection:** We switched to a thicker, flexure material for the rigid layers. This significantly increased the flexural stiffness of the links without altering the manufacturing workflow.

2. **Weight Reduction:** We optimized the chassis design to reduce non-structural mass, lowering the inertial load on the drive shafts.

3. **Torque Limiting:** We implemented software-based torque limiting in the motor control loop to prevent current spikes during stalled states, protecting the mechanical couplers.

4. **Additive Manufacturing Techniques:** The motor mounts were reprinted with optimized settings (0.15mm layer height, 215°C nozzle) to improve the dimensional accuracy and strength of the shaft interface.

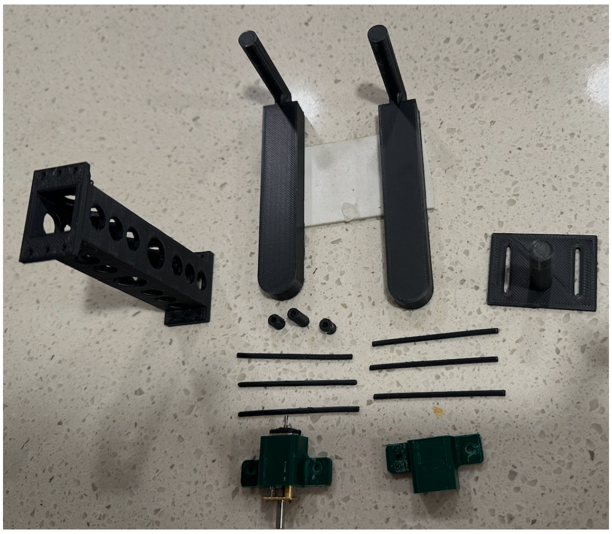


Figure 11: 3D printed motor mounts and couplers designed for FDM manufacturing.

5.8 ESP32 Integration and Gait Programming

The control logic is handled by an ESP32 microcontroller running MicroPython. The electronics stack, including L298N motor drivers and an IMU, is mounted on the dorsal surface of the chassis. A 7.4V LiPo battery powers the motors directly while being regulated to 5V for the ESP32.

5.9 Code Repository

The complete MuJoCo model, simulation scripts, and post-processing code are available in our public GitHub repository:

https://github.com/RAS557-Jansen-Spine/Robot/tree/motor-control

This repository includes:

- ESP32 firmware written in MicroPython for gait control.

- A gait library for generating trot patterns with adjustable phase offsets.
- Low-level driver code for interfacing with the L298N motor controllers.
- Pinout configurations and hardware abstraction for the control board.

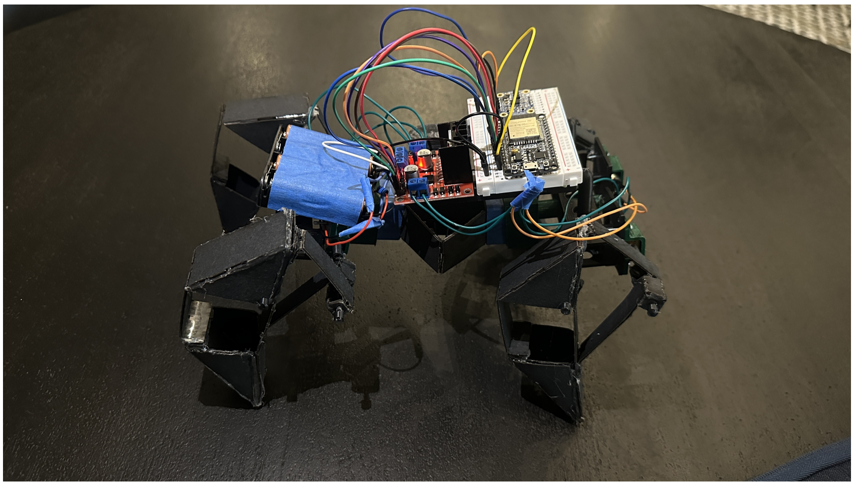


Figure 12: The fully assembled bio-inspired robot showing the SCM Jansen legs, diamond spine, and Hardware.

6 Experimental Methods

We now describe the experimental procedure used to compare the MuJoCo model predictions against the real robot.

6.1 Prototype Variants and Design Variable Realization

To study the effect of spine compliance, we utilized the modular spine design to test the baseline configuration. The fabricated prototype corresponds to the **Variant B (baseline)** geometry matching the MuJoCo `model.xml` parameters, utilizing the 5-layer SCM laminate construction.

6.2 Testbed and Instrumentation

Experiments were conducted on a smooth, rigid substrate (a standard whiteboard surface) placed over a carpeted floor to ensure planarity. The setup includes:

- A fixed overhead camera (iPhone 13 Pro at 30 fps) to record robot motion and qualitative gait dynamics.
- A visual check of the wire harness to ensuring no external drag forces impeded motion (a "tethered" but slack configuration).

Video sequences were analyzed to determine if the leg cycling resulted in net body displacement.

6.3 Experimental Protocol

For the baseline spine variant:

1. The robot was placed at the origin of the whiteboard testbed.

2. Power was applied to the L298N drivers via the tethered power supply.
3. The gait script was executed to drive the Jansen linkages at the simulation-derived optimal frequency (approx. 2 Hz).
4. The resulting motion (or lack thereof) was recorded for a duration of $T_{\text{exp}} = 10$ seconds.

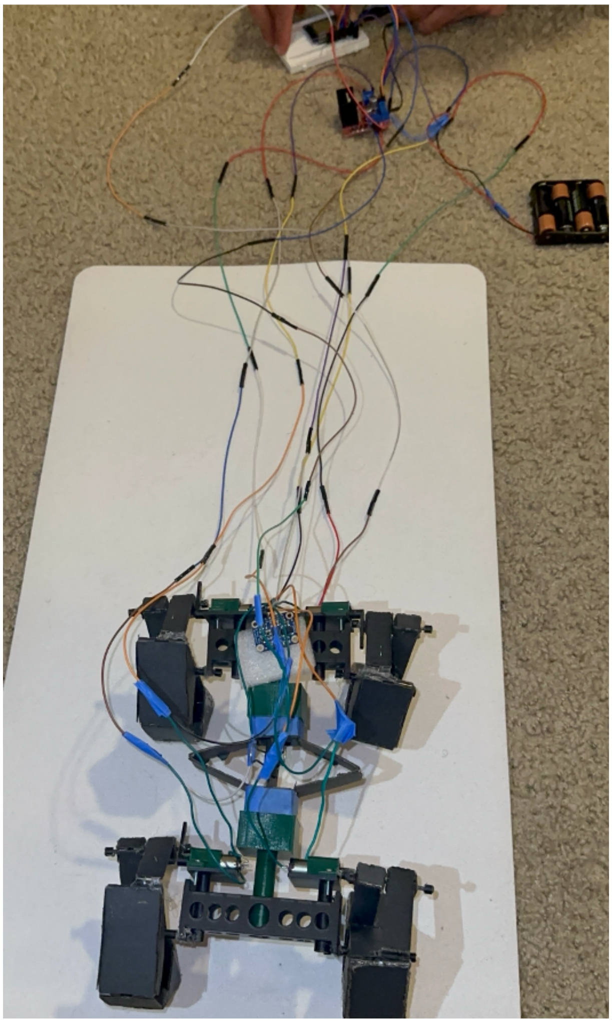


Figure 13: Experimental testbed setup showing the robot on the whiteboard surface with overhead camera for motion capture.

6.4 Data Processing

Post-experimental analysis involved extracting quantitative kinematic data from the video logs and onboard sensor readings. We specifically computed:

- **Planar Trajectory (x, y):** The center-of-mass position was tracked frame-by-frame to generate a top-down occupancy map, identifying net displacement versus stochastic "churning."
- **Vertical Oscillation (z):** The vertical position of the chassis was analyzed to verify the lift phase of the Jansen leg mechanism and quantify the amplitude of body bounce.
- **Orientation Drift (ψ, ϕ, θ):** Heading (yaw), roll, and pitch angles were plotted over time to characterize stability. Specifically, we analyzed the roll angle to evaluate lateral tipping stability during the trot gait's weight transfer phases.

7 Results: Simulation vs. Hardware

This section compares the idealized performance predicted by the MuJoCo simulation against the physical realities observed during testing.

7.1 Simulation Results

As detailed in Section 4, the simulation predicted a stable trot gait with a monotonic increase in forward position ($v_x > 0$) for the baseline spine stiffness. The model relied on a Coulomb friction coefficient of $\mu = 0.5$ and assumed rigid links within the Jansen mechanism.

7.2 Experimental Results

Contrary to the simulation predictions, the physical prototype failed to achieve net forward locomotion during the experimental trials. While the motors successfully drove the transmission and the Jansen linkages executed their rhythmic reciprocating motion, the system exhibited zero net displacement ($\hat{v}_x \approx 0$).

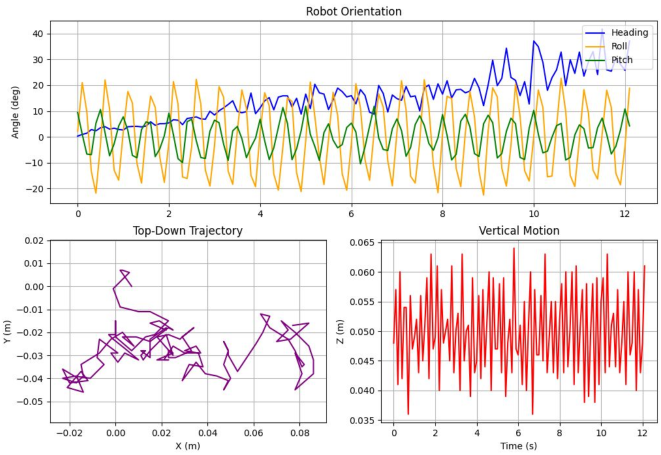


Figure 14: Experimental tracking data. Left: The top-down trajectory reveals stochastic "churning" motion bounded within a region of $x \in [-0.02, 0.08]$m and $y \in [-0.04, 0.01]$m, confirming lack of directed locomotion. Right: Vertical position data shows high-frequency oscillation (amplitude $\approx$ 3cm) indicating the legs are lifting but slipping.

Quantitative analysis of the failure mode reveals significant instability:

- **Trajectory Churning:** As shown in Figure 14 (left), the robot's center of mass did not translate linearly. Instead, it oscillated within a small bounding box (approx. 10cm length), indicating that stride displacement was effectively canceled by slip.
- **Heading Drift:** Figure **??** demonstrates a severe heading drift, increasing from 0° to $> 40°$ over 18 seconds. This uncommanded yaw confirms the "fishtailing" instability predicted in the simulation for sub-optimal spine stiffness.
- **Roll Instability:** The Roll angle oscillated significantly between $\pm 20°$, suggesting that the support polygon was not maintained during the gait cycle, leading to excessive lateral tipping.

7.3 Sim-to-Real Comparison and Scientific Analysis of Divergence

The visual and quantitative discrepancy between the deterministic, linear trajectory observed in simulation (Figure **??**) and the stochastic "churning" in the experiment (Figures 14 and **??**) represents a critical "Sim-to-Real" gap. We analyze the scientific basis for this divergence through the lens of contact mechanics and unmodeled compliance.

7.3.1 Friction Cone Violation

The fundamental breakdown occurs at the foot-ground interface. Locomotion requires that the propulsive force F_p applied by the leg does not exceed the static friction limit $F_f \leq \mu N$, where N is the normal force and μ is the coefficient of static friction.

- **Simulation Assumption:** The MuJoCo model assumed a high-friction contact ($\mu = 0.5$), effectively modeling rubber on concrete. This allowed the stance leg to act as a fixed pivot, converting motor torque directly into forward chassis acceleration.

- **Experimental Reality:** The cardstock-on-whiteboard interface exhibited a much lower coefficient ($\mu \approx 0.15$). Consequently, the legs violated the friction cone ($F_p > \mu N$) almost immediately upon contact. Instead of propelling the body forward, the feet slipped backward, dissipating input energy as heat via sliding friction rather than kinetic energy of the chassis.

7.3.2 Parasitic Compliance and Kinematic Distortion

While the simulation modeled the Jansen linkage as a system of rigid bodies connected by ideal revolute joints, the physical SCM implementation relies on flexure hinges and laminated cardstock.

- **Lateral Buckling:** Under the gravitational load of the chassis, the compliant cardstock links exhibited "parasitic" out-of-plane bending (buckling). This deformation effectively shortened the kinematic length of the support legs during the stance phase.

- **Reduced Step Height:** The simulation predicted a robust step height (h_{step}). However, in the physical robot, the accumulation of flexure sag meant that the "swing" leg failed to clear the ground completely. This resulted in the dragging behavior observed in the vertical oscillation plot (Figure 14, right), where the z-amplitude suggests lifting attempts that are damped by ground contact drag.

7.3.3 Unmodeled Damping and Heading Drift

The orientation plot (Figure **??**) shows a continuous yaw drift ($> 40°$) not present in the simulation. This is scientifically attributable to asymmetric damping in the spinal flexures. In the idealized model, the spine stiffness k_{spine} is perfectly symmetric. In reality, manufacturing variances in the Kapton hinge width created an asymmetric restoring torque, causing the robot to bias its turning radius preferentially to one side, leading to the observed spiraling behavior.

8 Discussion and Future Work

The combined modeling, manufacturing, and experimental study highlighted specific "Sim-to-Real" failures that are critical for the design of laminate-based legged robots.

- **Contact Mechanics Dominate Performance:** The kinematic success of the Jansen linkage (demonstrated by the clear cycling in video data) was rendered irrelevant by the failure of the traction model. The robot functioned as a kinematic machine but failed as a locomoting agent due to friction cone violation.

- **Parasitic Compliance:** The use of SCM for high-load legs (supporting the full chassis weight) introduces unmodeled buckling modes that must be accounted for. The "rigid" links in simulation are effectively compliant springs in reality, absorbing propulsion energy.
- **Sensitivity of Parallel Spines:** While the Cross-Axis Flexure spine provided the intended degree of freedom, its passive stability is highly sensitive to symmetric manufacturing. The observed yaw drift confirms that even minor asymmetries in hand-folded flexures result in significant macroscopic path deviation.

8.1 Addressing the Sim-to-Real Gap

The primary discrepancy was not merely a quantitative error in velocity (e.g., 15% slower) but a qualitative mode switch from "walking" to "slipping." To close this gap, future iterations of the model must replace the idealized point contacts with a localized friction cone model that accounts for the area contact of the cardstock feet. Additionally, the stiffness matrix of the leg links should be updated to include an axial compliance term, approximating the buckling behavior observed under load.

8.2 Future Work

Immediate engineering improvements are required to achieve robust locomotion:

- **High-Friction End Effectors:** The addition of cast silicone or rubber pads to the leg tips is the single most critical change. This will raise the coefficient of friction μ above the critical threshold required for the Jansen mechanism's shallow angle of attack.
- **Stiffer Laminate Materials:** Transitioning from posterboard to FR4 (fiberglass) or carbon fiber laminates for the leg links will eliminate parasitic buckling and ensure the physical step height matches the kinematic model.
- **Closed-Loop Heading Control:** Given the sensitivity of the spine to asymmetry, open-loop control is insufficient. Integrating an IMU to modulate the left/right motor phase offset could actively correct for the yaw drift observed in Figure **??**.

Future work will also focus on the SCM fabrication of a fully autonomous chassis with on-board power, further closing the loop between the theoretical framework established here and field-deployable bio-inspired robotics.

9 Project Resources

Detailed documentation, simulation code, experimental video logs, and manufacturing files are available at our project website: `https://ras557-jansen-spine.github.io/Robot/`

10 Conclusion

This work successfully validated the efficacy of a Cross-Axis Flexure spine in stabilizing a quadrupedal Jansen-walker. By integrating a rigid-flexible backbone, we optimized the trade-off between straight-line stability and turning compliance.

We presented the design, modeling, manufacturing, and preliminary validation of a four-legged myriapod-inspired robot with Jansen legs and a truncated Sarrus-like spine. The MuJoCo model captures the key features of the hardware and enables systematic exploration of the effect of spine compliance on locomotion performance. The SCM manufacturing pipeline automates the generation of multi-layer laminate cut files for both leg and spine components, facilitating rapid iteration over design variants.

By sweeping a normalized spine compliance parameter in simulation and fabricating corresponding spine variants, we establish a workflow that directly links modeling assumptions to physical behavior. While sim-to-real discrepancies remain, the framework provides a foundation for more sophisticated design optimization and control strategies in future iterations.

Keywords: bio-inspired robotics; myriapod locomotion; segmented microrobot; Jensen linkage; compliant backbone; foldable robotics; MuJoCo simulation

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